SPA617 Lab

RF Components: Characterization and Measurement

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1 RF Components: Characterization and Measurement

1.1 Aim

To understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

1.2 Objectives

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or passband.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(e) To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.

1.3 Theory

1.3.1 RF components in an RF receiver chain

A minimal RF receiver chain is shown in Fig. 1. A filter selects or rejects specified frequency ranges. A mixer translates a signal from one frequency to another. A low-noise amplifier (LNA) amplifies weak received signals while adding as little noise as possible. A power splitter/combiner distributes a signal to multiple paths or combines signals from different paths.

1.3.2 Filter transmission characteristics

A filter passes selected frequency components and attenuates others. The four commonly used filter types are:

- **Low-pass filter:** passes frequencies below its cutoff frequency and attenuates higher frequencies.
- **High-pass filter:** passes frequencies above its cutoff frequency and attenuates lower frequencies.
- **Band-pass filter:** passes frequencies between its lower and upper cutoff frequencies.
- **Band-reject or notch filter:** attenuates frequencies within a specified stopband and passes frequencies outside it.

For low-pass and high-pass filters, f_c denotes the cutoff frequency and f_s denotes the stopband edge. For band-pass and notch filters, f_1 and f_2 denote the lower and upper cutoff frequencies, while f_0 denotes the centre frequency or notch frequency.

For the transmission measurement, let P_{in} be the power measured without the filter and P_{out} be the power measured after inserting the filter. The insertion loss is

$$IL = P_{in} - P_{out} \quad \text{dB}.$$

A smaller value of IL indicates better transmission through the filter. The -3 dB cutoff frequencies are measured relative to the minimum insertion loss in the passband.

1.3.3 Filter reflection and return loss

When the input impedance of the filter, Z_{in}, is not equal to the characteristic impedance of the RF system, Z_0, a part of the incident signal is reflected from the filter input.

The voltage reflection coefficient is defined as

$$\Gamma = \frac{V_{\text{ref}}}{\text{inc}} = \frac{Z_{in} - Z_0}{}$$

V , $Z_{in} + Z_0$

where V_{inc} and V_{ref} are the incident and reflected voltage amplitudes, respectively.

The magnitude $|\Gamma|$ indicates the fraction of the incident voltage amplitude that is reflected. For a perfectly matched filter, $Z_{in} = Z_0$ and $|\Gamma| = 0$. For complete reflection, $|\Gamma| = 1$. The return

loss is

$$RL = -20\log_{10} |\Gamma| \quad \text{dB}.$$

A well-matched filter has a small reflection coefficient and a large return loss. A poorly matched filter has a larger reflection coefficient and a smaller return loss.

The important limiting cases are

Condition	$\lvert\Gamma\rvert$	RL
Perfect match	0	∞
Partial mismatch	$0 < \lvert\Gamma\rvert < 1$	> 0 dB
Complete reflection	1	0 dB

1.3.4 Quarter-wave open-circuit stub filter

An open-ended coaxial stub connected in shunt through a Tee connector behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction. The signal is therefore strongly attenuated, producing a notch.

The physical length of the stub is approximately

$$L = \frac{\lambda}{4} = \frac{VF \cdot c}{4 f_0},$$

where L is the stub length, λ is the guided wavelength in the cable, VF is the velocity factor, c is the speed of light in vacuum, and f_0 is the expected notch frequency. The connector length and open-end effect may cause the measured value of f_0 to differ slightly from the calculated value.

1.3.5 Mixer

An RF mixer is a three-port nonlinear device with RF, local-oscillator (LO), and intermediatefrequency (IF) ports. When signals at f_{RF} and f_{LO} are applied, the mixer produces the sum and difference components

$$f_{IF} = f_{RF} + f_{LO}$$

and

$$f_{IF} = f_{RF} - f_{LO} \quad ,$$

where the second expression assumes $f_{RF} > f_{LO}$. In a receiver, the required IF component is selected using a filter.

If P_{RF} is the RF input power and P_{IF} is the measured power of the selected IF component, the conversion loss is

$$\text{Conversion loss} = P_{RF} - P_{IF} \quad \text{dB}.$$

1.3.6 Power splitter/combiner

A power splitter divides the input power between two or more output ports. For an ideal twoway equal power splitter, each output is 3 dB below the input. A practical splitter has a small additional insertion loss.

Let P_{in} be the input power and let P_{out1} and P_{out2} be the powers measured at the two output ports. The losses to the two output ports are

$$\text{Loss}_1 = P_{in} - P_{out1}$$

and

$$\text{Loss}_2 = P_{in} - P_{out2} \quad .$$

A power combiner performs the reverse operation. The output depends on the frequencies, amplitudes, and relative phases of the two input signals.

1.3.7 RF amplifier gain and harmonics

An RF amplifier increases the power of an input signal. Its small-signal gain is

$$\text{Gain} = P_{out} - P_{in} \quad .$$

A nonlinear amplifier may be represented by

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots .$$

For

$$x(t) = A\cos(\omega t),$$

the output contains a DC component, the fundamental frequency ω, and harmonics at 2ω, 3ω, and higher frequencies:

$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos(\omega t) + \frac{a_2 A^2}{2}\cos(2\omega t) + \frac{a_3 A^3}{4}\cos(3\omega t) + \cdots .$$

At sufficiently high input power, the amplifier gain decreases. The 1-dB compression point is the value of P_{in} at which the gain has fallen by 1 dB from the small-signal gain. The corresponding amplifier output is denoted by P_{out}.

1.3.8 Third-order intercept (TOI)

When two input tones at f_1 and f_2 are applied to a nonlinear amplifier, the output contains the fundamentals f_1 and f_2, second-order products, harmonics, and third-order intermodulation products. The two third-order products closest to the fundamentals are

$$2f_1 - f_2$$

and

$$2f_2 - f_1.$$

The fundamental output increases with a slope of approximately $1:1$, while the IM3 output increases with a slope of approximately $3:1$. Their extrapolated intersection is called the third-order intercept (TOI), as shown in Fig. 5. In practice, the amplifier normally reaches compression before this theoretical intersection is reached.

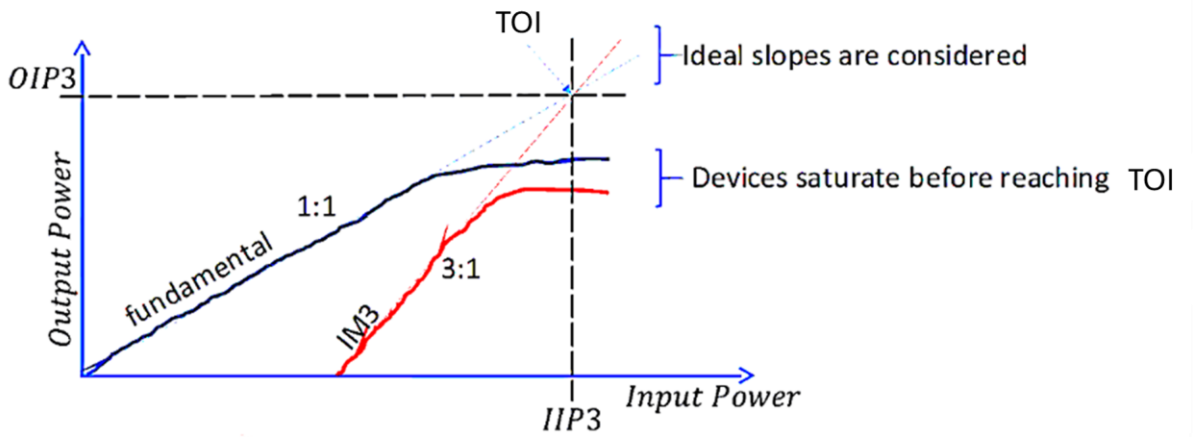


Figure 5. Third-Order Intercept – TOI (image source: Rahsoft.com).

1.4 Components Required

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- Filter
- Mixer
- Power splitter/combiner
- RF amplifier and the required DC power supply
- RF Tee connector
- Open-ended coaxial stub
- Short reference coaxial cable

- $50\ \Omega$ terminations
- Suitable RF cables, adapters, and attenuators

1.6 Experimental Setup

1.6.1 Transmission measurements

Setup A in Fig. 6 is used for filter insertion loss, mixer conversion loss, power-splitter/combiner measurements, and amplifier characterization. The RF signal generator is connected to the device under test, and the output is measured using the spectrum analyser.

1.6.2 Filter return-loss measurement

Setup B in Fig. 6 is used for the filter return-loss measurement. Connect the RF signal generator to one port of the Tee connector and the oscilloscope to the second port. Connect the filter input to the third port of the Tee connector.

Terminate the output port of the filter with a $50\ \Omega$ load so that reflections from the filter output are minimized. Set the oscilloscope input impedance to $1\ \mathrm{M}\Omega$ so that the oscilloscope does not significantly load the Tee junction.

Use the same cables and adapters throughout the measurement. The frequency sweep should be restricted to a small range within a relatively flat part of the filter passband. The Tee-connector method provides an approximate return-loss measurement because cable loss, connector mismatch, and oscilloscope loading are not removed by calibration.

1.7Experimental Procedures , Observations and Results

1.7.1 Part A: Filter characterization

A1. Transmission response and insertion loss

1. Connect the signal generator to the spectrum analyser using a short reference cable and the required adapters.
2. Set a suitable constant generator power.
3. Without connecting the filter, measure the power P_{in} at each test frequency.
4. Insert the filter without changing the generator setting, cables, or adapters.
5. Measure the output power P_{out} at the same frequencies.

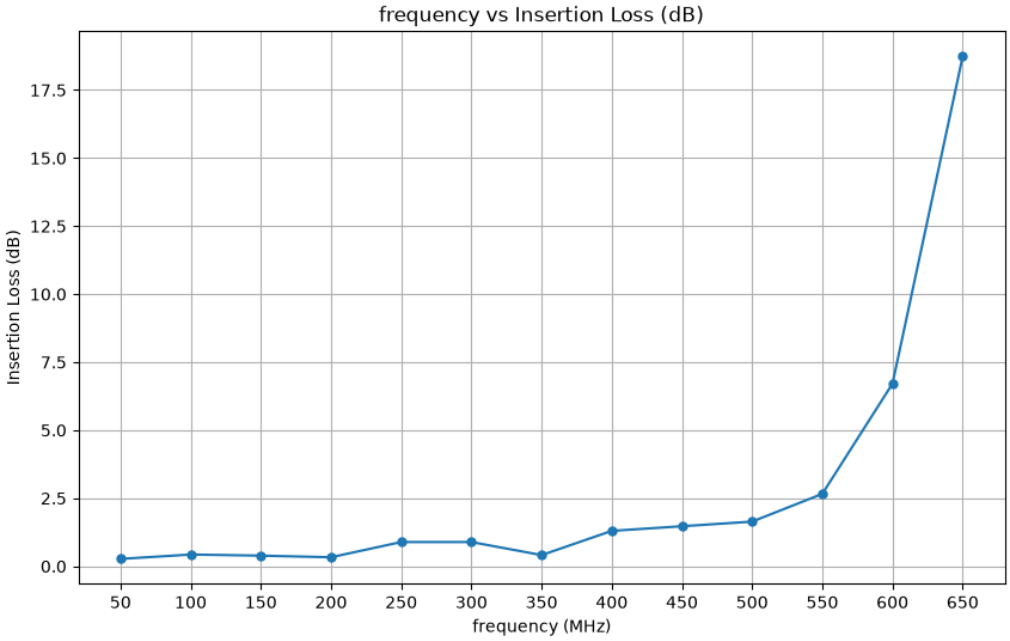
6. Calculate

$$IL = P_{in} - P_{out}.$$

7. Plot insertion loss versus frequency.

8. Identify the filter type. From Fig. 2, determine the relevant frequencies: f_c and f_s for a low-pass or high-pass filter, or f_1, f_0, and f_2 for a band-pass or notch filter.

Input frequency(MHz)	P_{in} without filter (dBm)	P_{out} with filter (dBm)	IL (dB)
50	-0.91	-1.18	0.27
100	-0.80	-1.23	0.43
150	-0.92	-1.31	0.39
200	-1.09	-1.42	0.33
250	-0.95	-1.84	0.89
300	-1.17	-2.06	0.89
350	-1.37	-1.78	0.41
400	-1.12	-2.42	1.3
450	-1.22	-2.69	1.47
500	-1.53	-3.17	1.64
550	-1.43	-4.09	2.66
600	-1.42	-8.13	6.71
650	-1.44	-20.21	18.73



Filter used is Low Pass Filter . Cutoff frequency for this filter is 400 MHz.

A2. Filter input reflection coefficient and return loss*

1. Make the connections shown in Fig. 6B.
2. Set the oscilloscope input impedance to $1\ \mathrm{M}\Omega$.
3. Select a small frequency range within a relatively flat portion of the filter passband.
4. Keep the signal-generator amplitude constant and sweep the frequency slowly.
5. Record the maximum voltage amplitude $V_{\max}$ and the adjacent minimum voltage amplitude $V_{\min}$ at the Tee junction.
6. Use the same voltage quantity for both readings. For example, if $V_{\max}$ is measured as a peak-to-peak voltage, $V_{\min}$ must also be measured as a peak-to-peak voltage.
7. Estimate the incident and reflected voltage amplitudes:

$$V_{\mathrm{inc}} = \frac{V_{\max} + V_{\min}}{2},$$
$$V_{\mathrm{ref}} = \frac{V_{\max} - V_{\min}}{2}.$$

8. Calculate the magnitude of the reflection coefficient:

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}.$$

9. Calculate the return loss:

$$RL = -20\log_{10}|\Gamma|.$$

Frequency range	$V_{\max}$ (mV)	$V_{\min}$ (mV)	V_{inc} (mV)	V_{ref} (V)	RL (dB)
139MHz – 261MHz	164	136	150	14	-20.63

Vinc = (164+136)/2 = 150 MHz

Vref = (164-136)/2 = 14MHz

$|\Gamma| = 14/150 = 0.093$ dB

$\text{RL} = -20\log_{10}|\Gamma| = -20^{*} = -20\log_{10} 0.093 = 20.63$ dB

A3. Quarter-wave open-circuit stub notch filter

1. Connect the signal generator to one port of a Tee connector and the spectrum analyser to the second port.
2. Connect an open-ended coaxial stub of length L to the third port.
3. Calculate the expected notch frequency:

$$f_0 = \frac{VF \cdot c}{4L}.$$

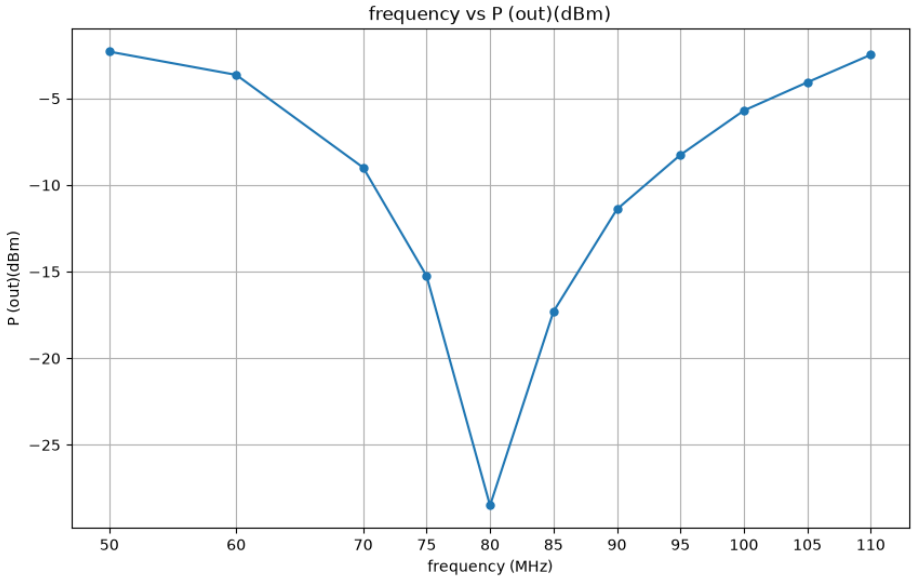
4. Without the stub, measure P_{in} over a frequency range around the expected f_0.
5. Connect the stub and measure P_{out} over the same frequency range.
6. Calculate the attenuation:

$$\text{Attenuation} = P_{in} - P_{out}.$$

7. Plot P_{out} versus frequency and identify the measured notch frequency f_0.

8. Compare the calculated and measured values of f_0 .

Frequency (MHz)	P_{in} without stub (dBm)	P_{out} with stub (dBm)	Attenuation (dB)
50	-0.71	-2.30	1.59
60	-0.72	-3.64	2.92
70	-0.50	-8.99	8.49
75	-0.48	-15.25	14.77
80	-0.50	-28.5	28
85	0.56	-17.30	16.74
90	-0.51	-11.40	10.89
95	-0.53	-8.26	7.73
100	-0.60	-5.70	5.1
105	-0.70	-4.07	3.37
110	-0.83	-2.48	1.65



The measured notch frequency f_0 = 80 MHz .

1.7.2 Part B: Mixer characterization

1. Check the working-frequency ranges and recommended power levels of the RF, LO, and IF ports.
2. Connect the RF and LO ports to two signal generators and connect the IF port to the spectrum analyser.
3. Apply input signals at f_{RF} and f_{LO}.
4. Calculate the expected values of

$$f_{IF} = f_{RF} + f_{LO}$$

and

$$f_{IF} = f_{RF} - f_{LO}.$$

5. Observe the output spectrum and identify the measured IF component.
6. Record the RF input power P_{RF} and the selected IF output power P_{IF}.
7. Calculate

$$\text{Conversion loss} = P_{RF} - P_{IF}.$$

8. Repeat for additional values of f_{RF} or f_{LO} if instructed.

f_{RF}	f_{LO}	Expected f_{IF}	Measured f_{IF}	P_{IF} (dBm)	P_{RF} (dBm)	Conversion loss (dB)
100 MHz	10 MHz	110 MHz	90 MHz	-9.62	-9.56	0.06
110 MHz	10 MHz	120 MHz	100 MHz	-9.56	-9.56	0.00
120 MHz	10 MHz	130 MHz	110 MHz	-9.48	-9.56	-0.08
130 MHz	10 MHz	140 MHz	120 MHz	-9.46	-9.28	0.18

1.7.3 Part C: Power splitter/combiner characterization

C1. Power splitter

1. Connect the splitter input to the signal generator.
2. Connect output port 1 to the spectrum analyser and terminate output port 2 with $50\ \Omega$.
3. Measure the input power P_{in} using the reference connection and then measure P_{out1}.
4. Interchange the output connections and measure P_{out2}.
5. Calculate

$$\text{Loss}_1 = P_{in} - P_{out1}$$

and

$$\text{Loss}_2 = P_{in} - P_{out2}.$$

6. Compare the measured power division with the ideal value of 3 dB at each output.

Frequency	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	Loss_1 (dB)	Loss_2 (dB)
50 MHz	-1.07	-4.42	-4.42	3.35	3.35

C2. Power combiner

1. Connect the two signal generators to the combiner input ports and connect the combined output to the spectrum analyser.
2. Set the two input signals to the assigned frequencies and safe power levels.
3. Observe and record the frequency components and power levels at the combined output.
4. Explain the observed output for equal-frequency and different-frequency inputs, as applicable.

f1 : frequency at input port 1 f2 : frequency at input port 2

P1 : Power delivered at input port 1 P2 : Power delivered at input port 2

P : power received at output

Same Frequency : f1 = f2 = 50 MHz at powers P1 = P2 = 0 dBm → Output Power P = -1.51 dBm

Distinct Frequencies : f1 = 50MHz , f2 = 10MHz at powers P1=-6.04dBm , P2=-1.6dBm

Then Output : Output has two peaks at 10MHz and at 50MHz with output powers -2.87dBm and 1.6dBm respectively.

Output power can differ from input power: practical losses, phase relationships, frequency response and measurement conditions affect the observed result.

1.7.4 Part D: Amplifier characterization

D1. Gain versus frequency :

1. Power the amplifier using the specified DC supply.
2. At a safe fixed generator setting, measure the amplifier input power P_{in}.
3. Insert the amplifier and measure the output power P_{out} over its operating band.

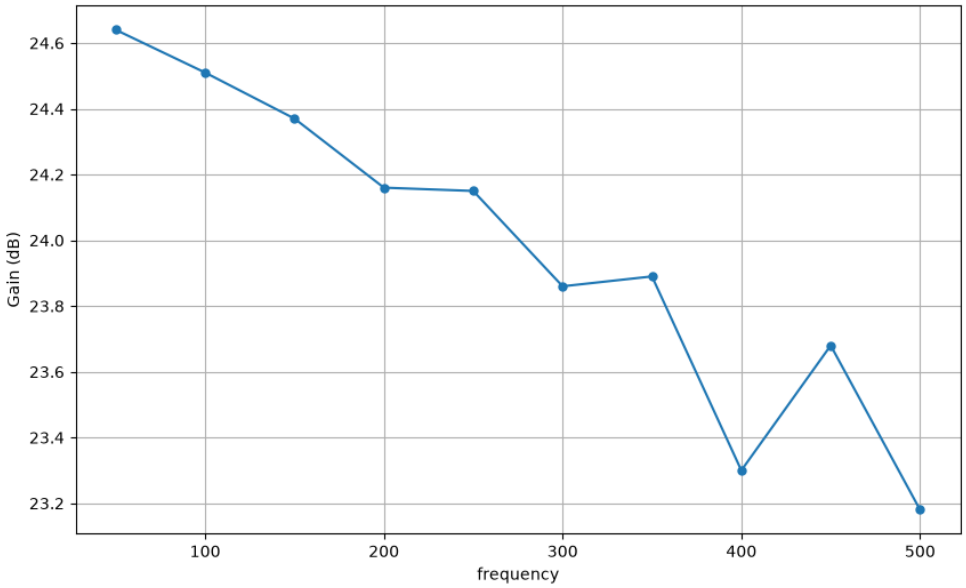
4. Calculate

$$\text{Gain} = P_{out} - P_{in}.$$

5. Plot gain versus frequency.

Frequency	P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
50	-20	4.64	24.64
100	-20	4.51	24.51
150	-20	4.37	24.37
200	-20	4.16	24.16
250	-20	4.15	24.15
300	-20	3.86	23.86
350	-20	3.89	23.89
400	-20	3.30	23.30
450	-20	3.68	23.68
500	-20	3.18	23.18

Plot of Gain(dB) versus frequency(MHz)



D2. Harmonics and 1-dB compression point

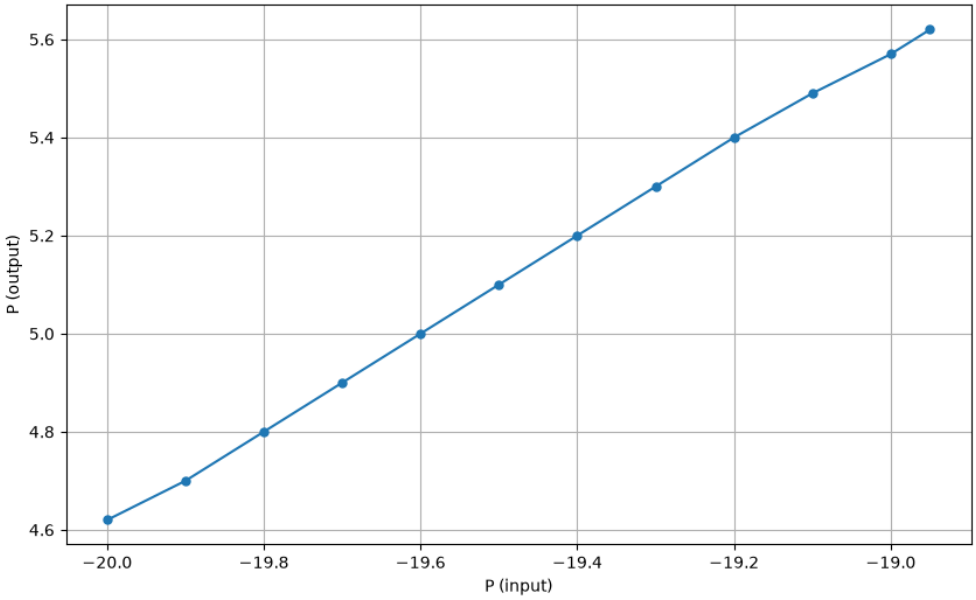
1. Select one frequency within the amplifier operating band.
2. Begin at a low value of P_{in} and record P_{out} and the visible harmonic power levels.
3. Increase P_{in} in safe steps without exceeding the amplifier or spectrum analyser limits.
4. At each step, calculate

$$\text{Gain} = P_{out} - P_{in}.$$

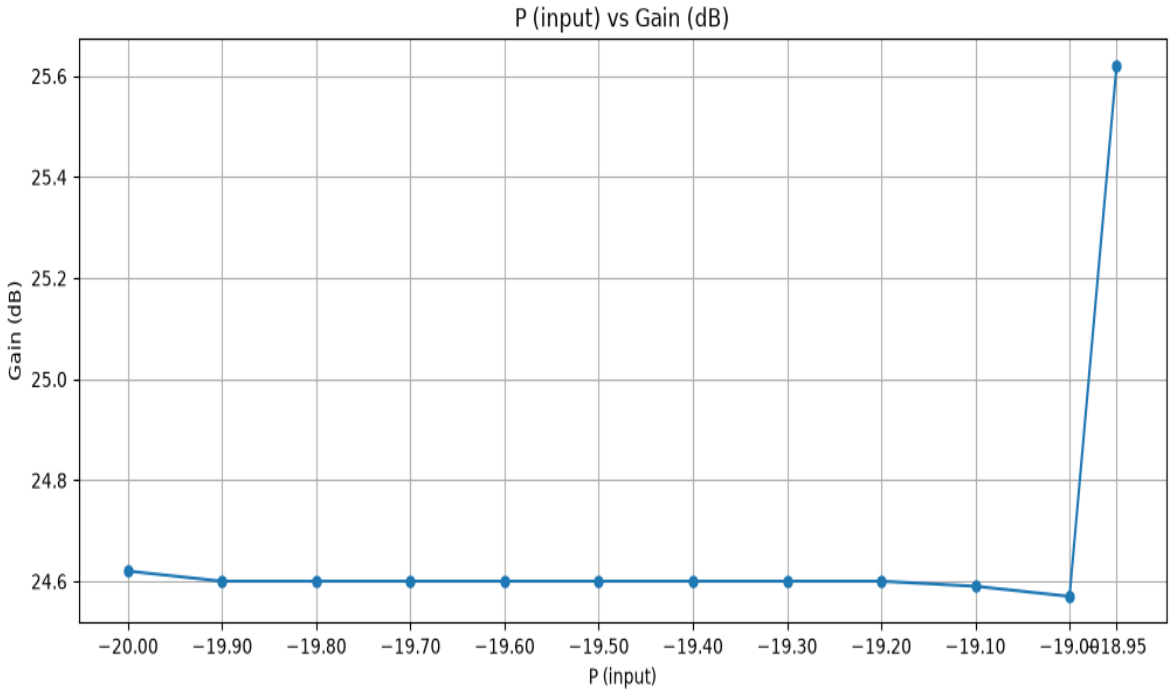
5. Plot P_{out} versus P_{in} and gain versus P_{in}.
6. Determine the values of P_{in} and P_{out} at which the gain has fallen by 1 dB from the small signal gain.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-20	4.62	24.62
-19.9	4.70	24.6
-19.8	4.80	24.6
-19.7	4.90	24.6
-19.6	5.00	24.6
-19.5	5.10	24.6
-19.4	5.20	24.6
-19.3	5.30	24.6
-19.2	5.40	24.6
-19.1	5.49	24.59
-19.0	5.57	24.57
-18.95	6.67	25.62

Plot of P_{out} versus P_{in}



Plot of Gain versus P_{in}



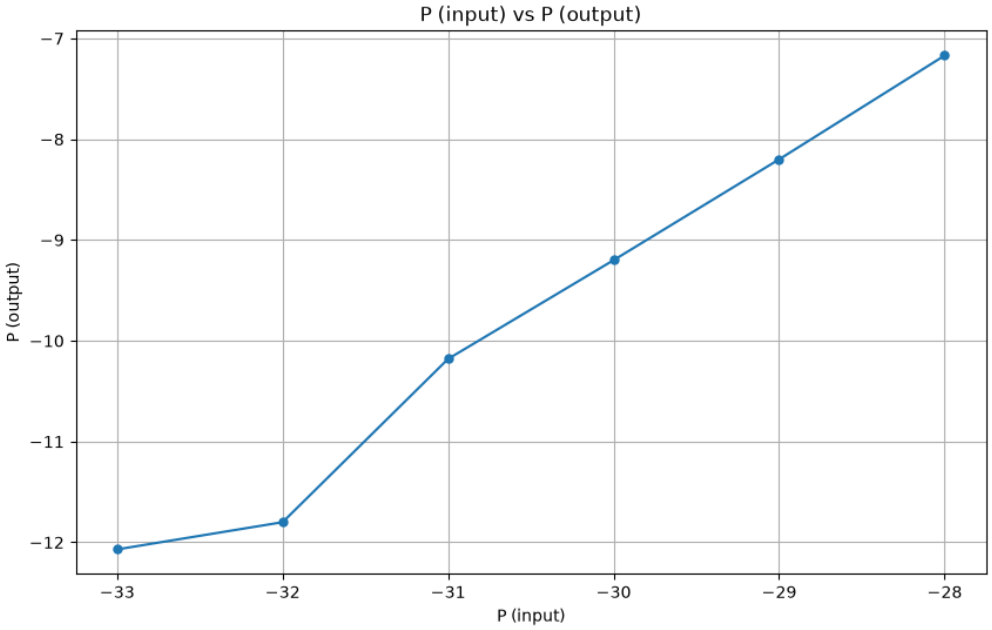
Started with P_{in} = -20 dB , P_{out} = 4.62dB and Gain = 24.62 dB

Finally 1dB compression happened at P_{in} = -18.95 dB , P_{out} = 6.67dB and Gain = 25.62 dB

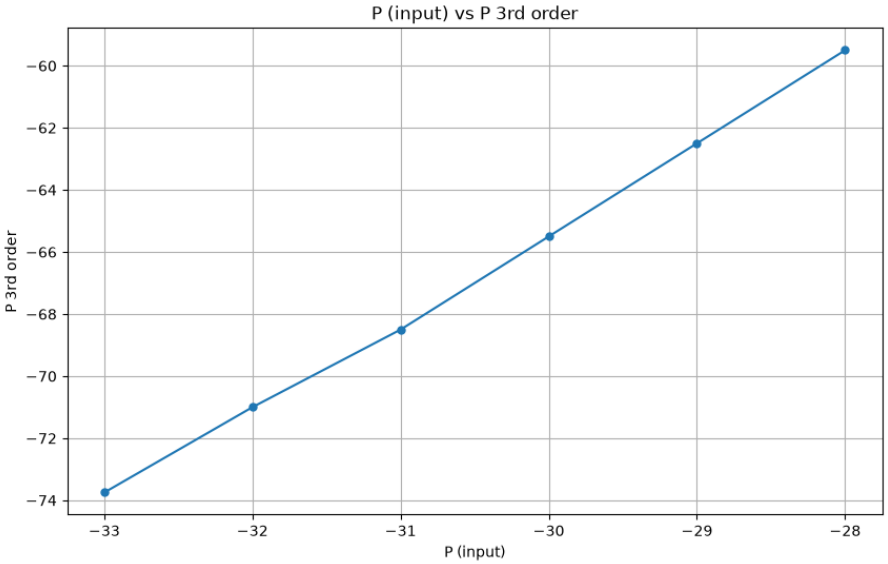
D3. Third-order intercept (TOI)

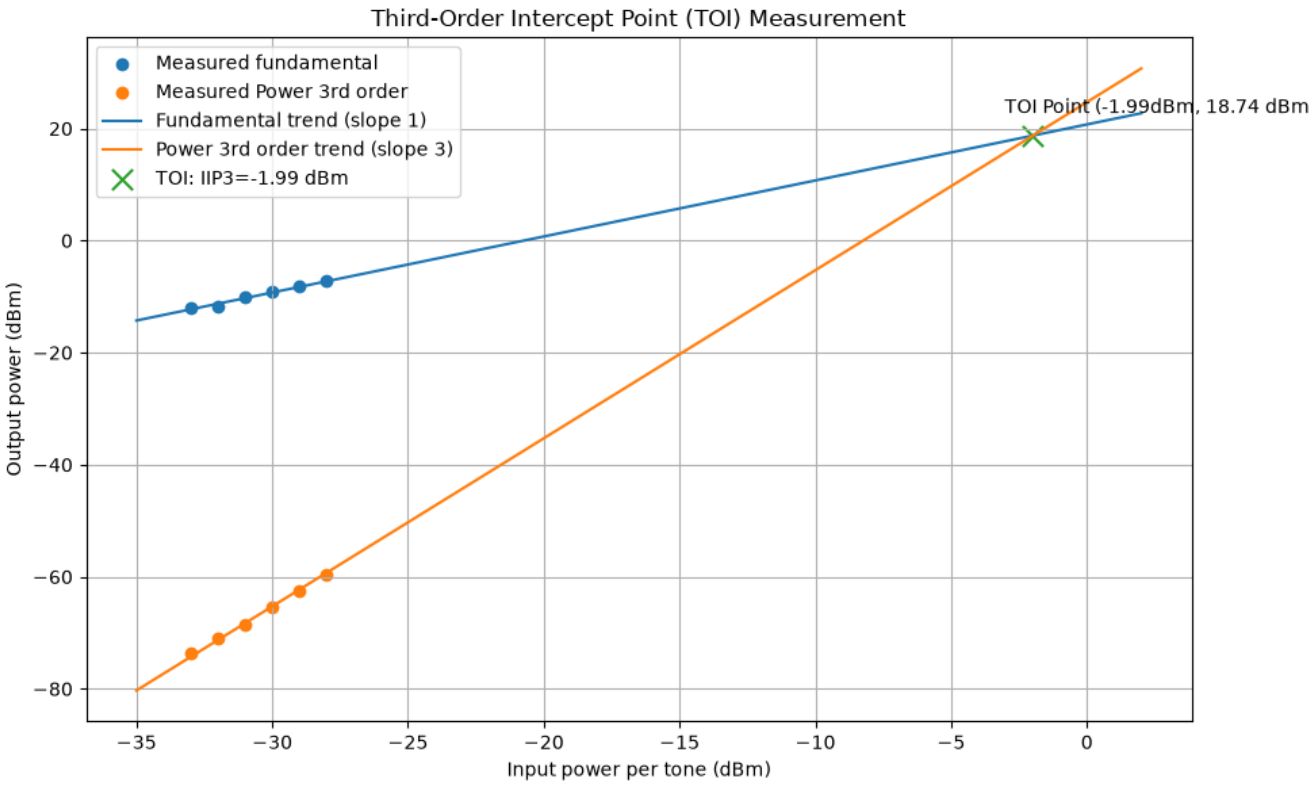
1. Set two closely spaced input frequencies f_1 and f_2 within the amplifier operating band.
2. Combine the two signal-generator outputs using the RF power combiner and apply the combined signal to the amplifier.
3. Connect the amplifier output to the spectrum analyser.
4. Identify the fundamentals at f_1 and f_2 and the third-order intermodulation products at $2f_1 - f_2$ and $2f_2 - f_1$.
5. For several safe values of input power per tone, record the fundamental output power P_{out} and the power $P_{3\text{rd-order}}$ of an adjacent third-order product.
6. Plot P_{out} and $P_{3\text{rd-order}}$ versus input power.
7. Extrapolate the fundamental and third-order trends using slopes of approximately $1:1$ and $3:1$, respectively.
8. Mark the intersection of the extrapolated lines as the TOI point.

P_{in} per tone (dBm)	P_{out} fundamental (dBm)	$P_{3\text{rd-order}}$ (dBm)
-33	-12.07	-73.75
-32	-11.8	-71.00
-31	-10.18	-68.5
-30	-9.2	-65.5
-29	-8.2	-62.5
-28	-7.17	-59.5



Plot of P_{in} vs $P_{3rd-order}$





Input intercept (IIP3) : **–1.99 dBm**

Output intercept (OIP3) : **18.74 dBm**

Measured fundamental slope : **1.037**

Measured IM3 slope : **2.850**

1.9 Conclusions

State the following results and briefly discuss their physical significance:

- the identified filter type, cutoff frequencies or passband, and insertion-loss response;

  The filter type was identified as Low Pass Filter from its frequency response as cutoff frequency is 400 MHz . The cutoff frequencies define the passband limits, while insertion loss indicates the reduction in signal power through the filter. On increasing frequency beyond 400 MHz Insertion Loss dramatically increased.

- the calculated and measured quarter-wave stub notch frequencies;

  The calculated and measured notch frequencies were compared above . At the notch frequency, the quarter-wave stub produces impedance transformation that causes strong signal rejection. quarter-wave stub notch frequency is 80 MHz .

- the observed mixer products and conversion loss;

  The mixer output contained the expected frequency components, including sum and difference frequencies, are noted above in the table. Conversion loss represents the difference between the input power at the selected input frequency and the output power of the desired converted component.

- the measured power division of the splitter/combiner;

  The measured output powers were measured and noted above and are used to evaluate the power division of the splitter/combiner. Ideally, an equal two-way splitter provides approximately 3 dB power division per output, excluding excess losses. Deviations from the ideal value indicate insertion loss, impedance mismatch and unequal power distribution.

- the amplifier gain, harmonics, 1-dB compression point, and third-order intercept point;

  The amplifier gain was determined from the difference between output and input power in dB. Harmonics were identified as unwanted frequency components at integer multiples of the input frequency. The 1-dB compression point represents the operating level at which gain falls 1 dB below its linear extrapolation. The third-order intercept point (TOI) was estimated by extrapolating the fundamental and third-order intermodulation trends. These parameters indicate the amplifier's gain, linearity and distortion performance.

- important sources of measurement uncertainty.

  Important sources of uncertainty include instrument calibration errors, frequency and power resolution, cable and connector losses, impedance mismatch, component tolerances,

environmental interference, signal-generator stability and reading errors. These factors can cause differences between theoretical and measured results